

Root Systems

1.1 Definition of Root Systems

Definition 1.1.1 ▶ s_v

Let E be a finite-dimensional real vector space endowed with an inner product written $(-, -)$. Given a non-zero vector $v \in E$, let s_v be the reflection in the hyperplane normal to v . Thus s_v sends v to $-v$ and fixes all elements y such that $(y, v) = 0$.

Proposition 1.1.2

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$$s_v(x) = x - \frac{2(x, v)}{(v, v)}v \quad \text{for all } x \in E$$

- s_v preserves the inner product, that is,

$$(s_v(x), s_v(y)) = (x, y) \quad \text{for all } x, y \in E.$$

- As a useful convention, we write

$$\langle x, v \rangle := \frac{2(x, v)}{(v, v)},$$

then

$$s_v = x - \langle x, v \rangle v$$

Remark.

The symbol $\langle x, v \rangle$ is only linear with respect to the first variable, x .

Definition 1.1.3 ▶ Root system

A subset R of a real vector space E is a **root system** if it satisfies the following axioms.

(R1) R is finite, it spans E , and it does not contain 0.

(R2) If $\alpha \in R$, then the only scalar multiples of α in R are $\pm\alpha$.

- (R3) If $\alpha \in R$, then the reflection s_α permutes the elements of R .
 (R4) If $\alpha, \beta \in R$, then $\langle \beta, \alpha \rangle \in \mathbb{Z}$.
 The elements of R are called **roots**.

1.2 First Steps in the Classification

Lemma 1.2.1 ► Finiteness Lemma

Suppose that R is a root system in the real inner-product space E . Let $\alpha, \beta \in R$ with $\beta \neq \pm\alpha$. Then

$$\langle \alpha, \beta \rangle \langle \beta, \alpha \rangle \in \{0, 1, 2, 3\}.$$

Proof sketch.

将 $\langle \alpha, \eta \rangle \langle \beta, \alpha \rangle$ 展开写成内积,

1.3 Basis for Root Systems

Definition 1.3.1

A subset B of R is a **base** for the root system R if

- (B1) B is a vector space basis for E , and
 (B2) every $\beta \in R$ can be written as $\beta = \sum_{\alpha \in B} k_\alpha \alpha$ with $k_\alpha \in \mathbb{Z}$, where all the non-zero coefficients k_α have the same sign.

Remark.

We say that a root $\beta \in R$ is **positive with respect to B** if the coefficients given in (B2) are positive, and otherwise it is **negative with respect to B** .